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Computer Science



Class 10

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Chapter 4

Notes

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Computer Science 10th
Chapter no 4
SLO Based Questions

Qno1: What is Data science. What are the main objectives of data science techniques?

Ans: Data Science refers to an interdisciplinary field of mathematics, statistics, and computer science.

Main objective of data science techniques is to get meaningful insight from the available data. The available data can be structured as well as unstructured. This objective is achieved through the steps of data collection, data cleaning, data preprocessing and data visualization. Various programming languages like Python, R studio and MATLAB can be used for data visualization.

Qno2: What is Artificial Intelligence?

Ans: The term **Artificial Intelligence** is referred to the ability of a machine to exhibit human behavior like problem solving, understanding natural language and interacting with the environment intelligently.

Qno3: What is a Recommender system?

Ans: A Recommender system is a tool or technology. It is used to recommend products, services or information to the user based on their past behavior and interests.

Qno4: Define the term Sentiment analysis.

Ans: Sentiment analysis is the term used to identify the sentiments of a customer by analyzing the review about the product. The sentiment can be positive, negative, or neutral. Sentiment analysis can be performed on reviews, text, opinions etc.

Qno5: What is Automation Industry?

Ans: Automation Industry: It is the use of computers and robots to perform industrial tasks automatically without human intervention. Examples are, robots assembling cars or electronic devices, automatic devices, automatic doors and lights etc

Qno6: What is NLP(Natural Language Processing)?

Ans: It is a subset of artificial intelligence, which enables the computers to understand, interpret and generate human language. For example, virtual assistants like Siri, Google, Assistant, Alexa and Cortana use NLP to understand human voice commands.

Qno7: What is Tensor Flow?

Ans: TensorFlow framework is tool that helps computer learn data like a robot. TensorFlow learns form data like a human being learn form experience.

Qno8: What is Deep Learning?

Ans: Deep learning is a type of computer learning which uses layers of artificial neural network to learn from data. It is a learning method which is implemented by using some tool like TensorFlow.

Qno9: What is Neural Network?

Ans: Neural network is also known as Artificial Neural Network. It is system designed to work like human brain. It is made up of artificial neurons. These tiny neurons make decisions like human brain.

Qno10: How many Machine Learning models are there?

Ans: Primarily, there are three models of machine learning:

1. Supervised Machine Learning Model
2. Unsupervised Machine Learning Model
3. Reinforcement Machine Learning Model

Qno11: What is Supervised Machine Learning Model

Ans: In this learning model a set of labeled data is provided to the machine/computer. The labeled data means that questions with their correct answers are provided to the machine. After reading/learning a lot of questions/answers the machine is trained. Now, only new questions without correct answers are provided to that machine and it provides us with the possible correct answers.

Qno12: What is Unsupervised Machine Learning Model

Ans: In unsupervised learning model, unlabeled data is provided to the machine/computer. The algorithm identifies the pattern in the data and divide it into groups based on similarities and differences. When the training is completed, the computer can identify new data and put it in the relevant category.

Qno13: What is Reinforcement Machine Learning Model

Ans: In reinforcement learning the machines are trained with the help of positive or negative feedback. To explain reinforcement learning we will use the following key terms:

1. Agent (machine/computer)

2. Environment (where agent takes actions)
3. Actions (moves made by agent)
4. Rewards (points for helpful actions)
5. Penalties (points deducted for unhelpful actions)

Qno14: What are the steps involved in Machine Learning Process?

Ans: Following is the brief description of Machine Learning Model:

Step 1: The preprocessed, cleaned data is split into Training data and Testing data. Sometimes we divide it into three parts and keep a small amount of data to validate the final results.

Step 2: A suitable algorithm is selected to train the machine. There are several algorithms like neural network, decision tree etc.

Step 3: The model is trained on the training set, it learns the pattern from data and develop a decision making ability.

Step 4: After the model is trained, test data and validation data is provided to the model and results are compared with actual values. The performance of the model can be improved by changing or improving the algorithm

Qno15: What are the similarities between Artificial Intelligence, Machine Learning and Data Science?

Ans:

Data Driven Approach:

All the three fields rely on data to provide optimal results. They use data to make predictions, recognize patterns and make decisions.

Use of Algorithm and Models:

All the three fields work through various algorithms. For example, artificial intelligence relies on search algorithms. Machine learning relies on supervised/unsupervised/reinforcement learning algorithms. Data science relies on statistical and visualization algorithms.

Goal of Automation:

The net goal of all the three fields is to automate various activities which were previously performed manually.

Qno16: What are the similarities between Artificial Intelligence, Machine Learning and Data Science?

Ans:

Tools and techniques:

They use various techniques, e.g. artificial intelligence uses neural networks, expert systems, and natural language processing.

Machine learning uses supervised learning, unsupervised learning, reinforcement learning and deep learning techniques.

Data science uses statistical analysis, data mining and data visualization techniques.

Application:

The application of artificial intelligence is in speech and image recognition, autonomous vehicles, chatbots and virtual assistants.

Application of machine learning is in recommendation system, fraud detection, predictive analysis, spam filtering, and personalized marketing.

Applications of data science includes business intelligence, market analysis, healthcare analytics, risk management and scientific research.

Output:

The output of artificial intelligence systems is that mimic human behavior.

The output of machine learning is prediction and classification.

The output of data science is insight, reports and visualization.

Qno17: What is Churn Prediction?

Ans: Churn Prediction is the technique to find out the possibility of a customer, to stop taking the services or products of a certain company. It is used in the fields of business and marketing. Churn prediction is needed to run the business successfully. It is often observed that getting new customers is less beneficial than keeping the existing customers loyal to your business.

Qno18: What is Data Visualization?

Ans: It is the process of creating graphical representation of data and information. It involves various visual elements like charts, graphs, and maps. This graphical representation is useful to make decisions in the field of science, arts and in daily life. It is a powerful tool that transforms raw data into meaningful and insightful information.

Qno19: What do you know about Data Visualization charts?**Ans:**

Following are different Data Visualization charts

Pie Chart: Scatter Plot: It is used to display the relationship between numerical variables. For example, correlation between height and weight of a group of people.

Histogram: It is used to show the distribution of a single numerical value. For example, distribution of score in a class.

Box Plot (Box-and-Whisker Plot): It is used to summarize the distribution of data with the help of minimum, first quartile, median, third quartile and maximum data. For example, analysis of salaries of various employees in a company.

Heatmap: It is used to represent data in matrix form. Data values are shown by using color intensity. For example, images received from space to analyze weather forecast.

Bubble Chart: It is similar to scatter plot, but it has a third variable which is represented by the size of bubble. For example, points in a 2D space where area of each point is proportional to the third variable.

Qno20: What are the different ways to represent data graphically?**Ans:**

There are different ways to represent data graphically to make it easier to understand. Some common types of data visualization are as follows:

Quantitative Visualization

It is used to represent numerical data. It focuses on quantities or numbers to show measurable data or the data that can be measured or counted. For example, a bar chart showing sales figures of a company over several months

Categorical Visualization

Categorical visualization is used to represent data that falls into distinct categories. This type is ideal for displaying nominal or ordinal data. For example, a pie chart showing the market share of different companies in a specific industry.

Temporal Visualization

Temporal visualization is used to display data that changes over time. It is used for time-series data. Line graphs are usually used to represent temporal data. For example, a line graph.

Spatial Visualization Spatial visualization is used to represent data related to physical locations or spaces, visualizing geographic or spatial data. For example, a heatmap showing population density across different regions.

Multivariate Visualization

Multivariate visualization is used to represent data involving more than two variables/dimensions. Scatter plots and heatmaps are effective tools to represent such data.

Interactive Visualization

Interactive visualization allows users to interact with the data, through digital platforms. The digital platforms can be dashboards and filters.

Statistical Visualization

Statistical visualization is used to present data to show statistical properties, such as distribution or correlation. Histograms, box plots, and scatter plots are popular choices.

Information Visualization

Information visualization is used to present complex data sets in an easier way, often for abstract or conceptual data. Network diagrams, tree maps, and word clouds are effective tools.

Qno21: What are the benefits/advantages of Data Visualization?**Ans:**

Data visualization has many advantages and benefits, some of them are as follows:

1. It helps non-technical stakeholders to understand complex data and make decisions regarding their business and services.
2. It helps the decision-makers understand the existing data and make well informed strategic decisions.
3. It improves the performance of related business or services.
4. It helps in better communication, understanding, and getting better insight from well-organized real-time data.
5. Data visualization saves time and identify trends quickly.

6. It saves time and enable us to make quick and accurate decisions as compared to manual analysis.

Qno22: What are the interrelated characteristics of Database and Machine Learning?

Ans:

Data storage	In database data is stored in structured format such as table.	Algorithms of machine learning require the same structured data to train their models. Datasets organized in rows and columns.
Scalability	DBMS can handle various sizes of databases ranges from student database of 50 students to the database of entire country or even more.	Machine learning requires scalable data storage solutions. In this way it can implement big data applications. Number of records in a dataset varies from hundreds to thousands and even in millions in a single dataset.
Realtime Data Processing	Database provides Realtime data storage and retrieval support.	Machine learning model requires Realtime data to train model for online data.
Analytical Tools	Database provides various tools for data analysis, e.g. SQL server and other software etc.	Database analytical tools are helpful for machine learning to provide fast and accurate solutions.

Qno23: What are the Data Visualization?

Ans:

Like Artificial Intelligence, data science and machine learning, data visualization is useful in almost all the fields of life. Some of them are as follows:

Business Intelligence: Data visualization helps to make data driven, well informed decisions. It is used to find market trends and helps to track and improve performance.

Healthcare: It helps to visualize the impact of various diseases affecting the patient. It is helpful to track disease and visualize the spread of disease.

Education: Data visualization is very helpful to teach data literacy skills, concept building, creative thinking, and critical thinking.

Science and Research: It is useful to visualize complex findings, very huge and complex data such as complex scientific data received from satellite in the form of photographs.

Sports and gaming: It is useful to visualize performance of players whether they are playing football in a ground or chess players playing in online tournaments. It is also helpful in sports broadcasting, and other predictions.

Finance: It is helpful to analyze market trends, to track portfolio performance and to identify investment opportunities.

Entertainment: It helps the entertainment industry, to visualize performance and to predict future trends. It helps in content optimization by visualizing audiences insights and trends.